

# John Yuehan Zhang

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## EDUCATION

### Sichuan University

Bachelor of Engineering in Cyber Science and Engineering

China

Sep 2023 – Jun 2027

- **GPA: 89.48/100**    **Rank: Top 10%**

### The Hong Kong University of Science and Technology (Guangzhou)

Visiting Student

China

Jul 2025 – Feb 2026

### University of California, Berkeley

Visiting Student

USA

Currently

## RESEARCH INTERESTS

Fault-tolerant quantum computing, quantum error correction, quantum computing architecture and compiler.

## PUBLICATIONS (\* Equal contribution)

- [1] **John Yuehan Zhang**, *CircLS: Compiling Lattice Surgery to Physical Circuits with Dynamic Allocation*, arXiv:2608.23819, 2026, (**Under review at ASPLOS 2027**). [\[paper\]](#) [\[code\]](#)
- [2] Keming He\*, **Yuehan Zhang\***, Hongshun Yao, Jin-Guo Liu, and Xin Wang, *Block Coordinate Descent for Dynamic Portfolio Optimization on Finite-Precision Coherent Ising Machines*, arXiv:2603.23200, 2026, (**Under review at Quantum Science and Technology**). [\[paper\]](#)

## RESEARCH EXPERIENCE

### CircLS: Compiling Lattice Surgery to Physical Circuits with Dynamic Allocation

- **Unified lowering algorithm:** Designed a linear-time algorithm that lowers any Pauli product measurement (PPM) sequence to a stim circuit through stabilizer construction rules, so that a compiled program can be verified at the circuit level and its logical error rate (LER) measured from real samples.
- **Dynamic patch allocation:** Proposed allocating each data patch at its first use and freeing it at its last use, with freed tiles reused as ancilla paths; built a physical-level compiler around it, including a mapper that keeps ancilla paths short and a router that draws each PPM's path through the free tiles.
- **Comprehensive evaluation:** Against two prior toolchains, reduced the allocated spacetime volume by 34% and 27%, the LER by 40% and 48%, and the compile time by 98% and 96%, and compiled programs that they cannot.

### Block Coordinate Descent for Dynamic Portfolio Optimization on Finite-Precision Coherent Ising Machines

- **QUBO modeling:** Formulated dynamic portfolio optimization as a QUBO problem, encoding expected returns, risk, transaction costs and budget constraints into one quadratic objective.
- **Scalability via BCD:** Proposed a block coordinate descent scheme that decomposes the QUBO along the time axis into bounded-size time-block subproblems with weak inter-block coupling, so that large instances fit the finite input precision of the hardware.
- **Quantum experiments:** Implemented the pipeline on a coherent Ising machine (CIM) and benchmarked it against classical solvers and gate-based quantum algorithms (e.g., QAOA), obtaining competitive portfolios with reduced runtime under hardware precision limits.

## HONORS AND AWARDS

- First Prize, 15<sup>th</sup> National College Student Mathematics Competition, Sichuan Division    Sep 2024
- First Prize, 5<sup>th</sup> MathorCup Mathematical Application Challenge - Big Data Competition    Dec 2024
- Honorable Mention, The Mathematical Contest in Modeling (MCM)    May 2025

## SKILLS

- **Programming Skills:** Python, Java, C, C++, LaTeX
- **Language Skills:** English (professional, TOEFL: 104), Chinese (native speaker)